

Machine Learning for Electrocatalyst Discovery in Energy Conversion

Abstract

Machine learning has become a central methodological layer in modern electrocatalyst discovery, sitting between first-principles modeling, high-throughput experimentation, and device-level validation. In energy-conversion catalysis, the main promise of machine learning is not merely faster regression on density functional theory outputs, but a reorganization of the discovery process itself: representations become more expressive, screening becomes uncertainty-aware, experimental campaigns become adaptive, and inverse design becomes feasible over search spaces that are too large for exhaustive exploration. This survey reviews machine learning for electrocatalyst discovery across hydrogen evolution, oxygen evolution, oxygen reduction, carbon dioxide reduction, and nitrogen reduction, with emphasis on methodological questions of interest to the artificial intelligence and computer science communities. The survey first frames electrocatalyst discovery as a family of prediction, ranking, optimization, and generation problems. It then analyzes the main method families: descriptor-based learning, graph and geometric learning, machine-learned interatomic potentials, active learning and Bayesian optimization, multimodal and language-assisted models, and emerging inverse-design frameworks. Across these families, the survey examines how model design interacts with catalyst representations, benchmark construction, data heterogeneity, uncertainty quantification, and electrochemical realism. It further synthesizes application patterns across single-atom catalysts, dual-atom motifs, high-entropy alloys, low-dimensional materials, oxides, and Cu-based multicarbon CO₂ reduction systems. The central conclusion is that the field has moved beyond proof-of-concept prediction toward integrated, closed-loop discovery systems, but remains constrained by domain shift between idealized calculations and operando conditions, by insufficiently standardized benchmarks for activity-selectivity-stability tradeoffs, and by limited treatment of electrolyte, potential, reconstruction, and degradation. Progress will likely depend on models that are simultaneously physically grounded, data-efficient, uncertainty-calibrated, multimodal, and experimentally interfaced.

Introduction

Electrocatalysis is a cornerstone of sustainable energy conversion because it underlies water electrolysis, fuel cells, rechargeable metal-air systems, nitrogen fixation, and electrochemical carbon utilization. The scientific basis for electrocatalyst discovery has long been shaped by adsorption-energy descriptors, scaling relations, free-energy diagrams, and volcano analyses, all of which made it possible to connect electronic structure to catalytic activity and to screen candidate materials computationally [1-8]. In parallel, the broader machine-learning literature in materials science and heterogeneous catalysis established the conceptual basis for data-driven prediction, representation learning, and inverse design [9-13]. What changed over the last several years is that electrocatalysis has become a sufficiently large and methodologically mature subfield to support dedicated reviews, benchmarks, and communities, rather than appearing only as an application domain inside general catalyst informatics. That maturation is visible in the emergence of substantial scholarly reviews from 2022 onward, including Chemical Reviews, Current

Opinion, and ACS Materials Au surveys devoted specifically to electrocatalysis and AI-enabled electrocatalyst discovery. ¹

Machine learning is attractive in electrocatalysis because the discovery problem is structurally difficult. Reactivity depends on composition, surface orientation, local coordination, adsorbate identity, coverage, solvation, applied potential, surface reconstruction, and often time-dependent restructuring. Even when the objective is narrowed to one elementary step, the relevant design space typically remains combinatorial: alloy compositions, dopant concentrations, support effects, active-site motifs, and catalyst morphology each multiply the search burden. Density functional theory provides mechanistic interpretability and tractable first-principles labels, but at a cost that becomes prohibitive for global searches or for explicit treatment of interfaces and dynamics [1-8,16-24]. Machine learning therefore enters at several levels: surrogate prediction of adsorption energies or overpotentials, ranking and triage of candidates for DFT, active learning for dataset construction, structure relaxation with learned force fields, and closed-loop optimization with experiments [15-24].

From the perspective of computer science, electrocatalyst discovery is especially interesting because it combines several challenging learning settings. The same program may require low-data learning from expensive labels, geometric deep learning over periodic atomistic structures, multi-objective optimization under uncertainty, multimodal fusion of graphs, spectra, and text, and domain adaptation from calculations to experiments. The modeling targets are also unusually diverse. Some studies predict scalar proxies such as adsorption energies, d-band-derived descriptors, free-energy changes, limiting potentials, or overpotentials [25-28,65-81]. Others predict relaxed structures, forces, transition states, adsorption-site placements, catalytic motifs, or experimental outcomes such as Faradaic efficiency and current density [29-42,61-64,117-131]. More recent work attempts to learn design policies rather than property values, using Bayesian optimization, reinforcement learning, diffusion models, or large language model-based proposal systems [151-162].

Another reason the field has advanced rapidly is the appearance of shared benchmark ecosystems. OC20 introduced a million-scale adsorption/relaxation benchmark for heterogeneous catalysis; OC22 extended benchmark design to oxide electrocatalysts; AdsorbML operationalized efficient global-minimum adsorption-energy search; OCx24 introduced experimental data explicitly intended to bridge AI and electrocatalysis; and OC25 moved benchmark development toward electrified solid-liquid interfaces. OC20 alone reports 1,281,040 DFT relaxations, while AdsorbML reports a balanced operating point that recovers the lowest-energy configuration 87.36% of the time with roughly 2000× speedup; OCx24 adds 572 synthesized samples and hundreds of gas-diffusion-electrode evaluations under industrially relevant current densities. ² These resources have changed what it means to make progress: model comparisons can now be benchmarked more systematically, out-of-distribution generalization is tested more explicitly, and data scale is large enough to motivate architectural innovation rather than small-dataset curve fitting.

This survey treats machine learning for electrocatalyst discovery as an academic field in its own right. The aim is not only to summarize representative applications, but also to extract the methodological structure of the area. The discussion is therefore organized around problem formulation, data ecosystems, representation choices, model families, closed-loop optimization, and reaction-specific discovery patterns. Throughout, the focus remains on energy-conversion electrocatalysis: hydrogen evolution reaction (HER), oxygen evolution reaction (OER), oxygen reduction reaction (ORR), carbon dioxide reduction reaction (CO₂RR), and nitrogen reduction reaction (NRR), together with catalyst classes such as single-atom catalysts,

dual-atom catalysts, high-entropy alloys, complex oxides, and low-dimensional materials [1-24,35-42,85-150].

Methods

Problem formulations and data ecosystems

Most machine-learning workflows in electrocatalysis can be cast into four problem classes. The first is **forward property prediction**, where the model estimates a scalar or vector target from a catalyst representation. Typical labels include adsorption energies, reaction free energies, overpotentials, stability metrics, limiting potentials, surface energies, and in some cases experimentally measured performance quantities such as Faradaic efficiency, selectivity, onset potential, or durability [1-8,25-28,65-84]. The second is **ranking and screening**, where the objective is not minimal mean absolute error per se, but reliable prioritization of top candidates under a fixed computational budget [25-27,73-76,85-145]. The third is **structure or state prediction**, such as adsorption-site placement, relaxed geometry prediction, force prediction, or transition-state estimation [29-34,41-42,43-64]. The fourth is **inverse design and sequential decision making**, where the model proposes new catalysts or new experiments to maximize an acquisition function, satisfy design constraints, or trade off multiple objectives [25,82,106-108,123-126,143-145,151-162,196].

Data sources in these workflows span three broad regimes. One regime is **small, curated DFT datasets**, common in early descriptor-based studies of HER, OER, CO₂RR, and NRR, where models are trained on tens to a few hundreds of catalysts and descriptor engineering dominates overall performance [25-28,65-81,85-116,132-150]. A second regime is **large atomistic benchmark datasets**, especially OC20, OC22, AdsorbML-derived dense adsorption sets, and more recent extensions such as AQC_{at}25 and OC25, which support geometric deep learning and machine-learned force-field development [29-34,41-42,43-64,174-175]. A third regime is **experimental or hybrid datasets**, still much smaller but increasingly important for realistic discovery, including oxygen-evolution composition scans, synchrotron-characterized catalyst libraries, and the OC_x24 calculation-to-experiment bridge [33,74,104-110,142-145,194-196]. The shift from low-data descriptor fitting to benchmark-centered learning has been one of the most important structural changes in the field. ³

Benchmark choice strongly shapes model behavior. If the task is adsorption-energy prediction on idealized slabs, then the model may exploit compositional or local-geometric regularities that do not survive transfer to reconstructed, solvated, or potential-dependent surfaces. If the task is relaxed-energy prediction on OC20, then success depends on geometry sensitivity and generalization across element combinations and adsorbates. If the task is candidate ranking in a discovery loop, then uncertainty calibration and top-k hit rate can be more important than aggregate regression error. This means that electrocatalysis is not a single benchmark problem but a nested sequence of problems, from local surrogate accuracy to chemically meaningful prioritization to experimentally validated design [25-34,38-42,75-76].

Representations and descriptors

Representation design remains a central methodological axis because electrocatalytic activity is inherently local but not purely local. The simplest representations treat the catalyst as a vector of elemental, structural, and electronic descriptors. Examples include electronegativity, valence-electron count, ionic radius, d-band proxies, coordination numbers, adsorption-site counts, or manually engineered local-environment statistics

[27,65-81,183-184]. Such representations are especially effective when datasets are small and when the scientific goal includes interpretability. They often reveal meaningful structure-activity relationships, identify dominant features, and support physically interpretable screening rules for single-atom catalysts, alloy surfaces, transition-metal hydroxides, and dual-atom motifs [67-81,103-116,118-150].

Descriptor-based learning, however, has intrinsic limits. It usually presumes that the important chemistry has already been compressed into a fixed feature vector, and its success often depends on knowing in advance which geometric or electronic quantities matter. That assumption becomes fragile for multimetallic catalysts, high-entropy alloys, complex oxides, and heterogeneous interfaces where local environment determines behavior in ways that are difficult to summarize manually [35-36,73,77,138-145,178-180]. This is one reason why the field moved toward graph- and geometry-based representations.

In graph-based representations, atoms are nodes, edges encode local neighborhoods or interatomic distances, and message-passing updates learn latent features from atomic composition and geometry. For surface catalysis, the graph often includes both slab atoms and adsorbates, allowing the model to encode catalyst-adsorbate interactions directly [43-64]. Directional and equivariant models improved this line of work by incorporating angles, higher-order geometric interactions, and rotational symmetries. DimeNet, GemNet, PaiNN-like architectures, MACE, eSCN, and Equiformer-class models illustrate a broader transition from invariant atomistic regressors to symmetry-aware geometric learners [46-57]. In electrocatalysis this matters because adsorption energies and forces are highly sensitive to site geometry, adsorbate orientation, and reconstruction pathways, and those effects are not well captured by purely composition-based descriptors [29-34,41-42,58-64].

A newer representational direction is **multimodality**. Text-based models such as CatBERTa linearize atomic or site information into token sequences [61]. Multimodal transformers such as AdsMT combine graph encoders with adsorbate feature vectors to avoid explicit enumeration of adsorption sites [62]. Language-and-graph pretraining attempts to share latent spaces between atomistic structures and textual encodings [64]. At the experimental end, multimodal descriptors derived from XANES, EXAFS, XRD, SAXS, and photoelectron spectroscopy have been used to connect structural characterization to catalytic performance and to learned discovery rules [194]. These developments suggest that future electrocatalyst representations will not be purely atomic or purely tabular. Instead, they will likely integrate local structure, composition, electronic descriptors, processing history, characterization signatures, and literature-derived priors.

Predictive models

The earliest and still widely used electrocatalysis models rely on classical supervised learning: linear regression, support vector methods, random forests, Gaussian processes, gradient boosting, XGBoost, kernel ridge regression, and ensemble methods [25-28,65-81]. These methods are often competitive when datasets are small, descriptors are physically meaningful, and the desired output is a scalar target. Their advantages are clear. They can be trained inexpensively, uncertainty can sometimes be estimated well, and feature attributions can be extracted with relatively low effort. In HER and OER studies based on MXenes, MBenes, hydroxides, or single-atom motifs, such models have repeatedly accelerated discovery relative to brute-force DFT screening [78-81,88-90,98,103-105].

Gaussian-process and tree-based surrogates have been especially influential in sequential discovery loops because they naturally support uncertainty-aware acquisition. The active-learning strategy of Tran and Ulissi

on intermetallics and the Nature study by Zhong et al. on Cu-Al CO₂RR catalysts are canonical demonstrations of this paradigm [25,117]. Here, the point is not just interpolation. The model allocates new DFT calls adaptively, balancing exploration and exploitation, and thereby turns computational screening into a design loop. Similar ideas now appear in multimetal OER optimization, high-entropy oxide exploration, and Bayesian optimization over quaternary composition spaces [106-108,142-145,196].

Deep learning becomes decisively advantageous when one moves from scalar descriptors to atomistic structures. Crystal graph convolutional networks, SchNet, and later directional and equivariant models were important because they removed the need for extensive hand-crafted feature engineering [43-55]. On electrocatalysis benchmarks, geometric models now power force prediction, structure relaxation, adsorption-energy estimation, and out-of-distribution generalization across new adsorbates or catalyst compositions [29-34,41-42,56-64]. This direction is not merely about better test error. It changes the scale of feasible computation by enabling learned relaxations and by amortizing DFT-like information over large candidate spaces.

Machine-learned interatomic potentials are particularly important. In traditional workflows, candidate enumeration is often bottlenecked by geometry optimization, not just by single-point evaluation. AdsorbML showed that pretrained machine-learned force fields can substantially reduce the cost of locating low-energy adsorption structures, while CatTSunami extended pretrained geometric models toward transition-state energy estimation [32,42]. The latest datasets push further toward realistic interfaces: OC25 explicitly targets electrochemical reactions at electrified solid-liquid interfaces, where explicit solvent and interfacial complexity had previously limited benchmark development. ⁴

Interpretability remains an active concern even as models grow more expressive. Several papers respond by learning sparse or physically meaningful descriptors, using local chemical-environment explanations, subgroup discovery, forward feature selection, game-theoretic attribution, or interpretable dual-atom motif models [68-72,79-81,183-185]. The important insight is that interpretability in electrocatalysis is not only a human-centered desideratum. It is also a route to transfer: if a model can identify which site properties control activity or selectivity, then those regularities can generalize better across catalyst families than purely correlational latent embeddings.

Closed-loop optimization and inverse design

A major shift in recent work is from static prediction to **closed-loop discovery**. In this setting, a machine-learning model is embedded inside a sequential workflow that alternates prediction, acquisition, evaluation, and retraining. The evaluation can be DFT, higher-level simulation, high-throughput synthesis, or electrochemical testing [25,28,74,106-108,142-145,151-160,196]. This differs conceptually from conventional screening. Rather than exhaustively scoring a predefined candidate pool, the model actively shapes the data distribution it will subsequently learn from.

Bayesian optimization is the dominant framework in this area because electrocatalyst discovery is expensive and usually multi-objective. Researchers typically must trade off activity, selectivity, stability, abundance, cost, and sometimes synthesis feasibility. Acquisition functions can prioritize expected improvement, upper confidence bounds, and constrained optimization over these competing goals [25,108,143-145,153-160,196]. In electrocatalysis, Bayesian loops have been used for Cu-based CO₂RR optimization, multimetal OER catalyst discovery, alloy ordering in Pt-based fuel-cell catalysts, and compositional design in quaternary material systems [82,106-108,124,143-145,193,196].

Inverse design methods extend this logic by learning proposal distributions instead of or in addition to surrogates. Variational autoencoders, reinforcement-learning systems, GFlowNets, diffusion models, and more recent large-language-model-based proposal systems each instantiate a different answer to the same question: how should the search algorithm navigate a high-dimensional catalyst space under sparse feedback? [12-13,151-162]. For electrocatalysis, this question is not yet settled. Many inverse-design papers still depend on proxy objectives that are easier to model than the final electrochemical target. Yet the methodological movement is significant. AdsorbDiff addresses inverse placement of adsorbates on surfaces [63]. Topology-based variational autoencoders target controllable active-site generation [162]. Large language model systems have begun to appear as literature-aware or retrieval-augmented proposal engines [155,157]. Even where these methods remain preliminary, they signal a transition from “predict then rank” toward “generate, constrain, and optimize.”

Autonomy also requires bridging modeling and experimentation. Human-machine collaboration studies in OER, multimodal synchrotron-informed discovery, and OCx24-style experimental benchmark building are early examples of this integration [107,194]. The central technical difficulty is that experiments return noisier, more heterogeneous labels than DFT, while also embodying processing and transport phenomena that slab models ignore. Closed-loop electrocatalysis therefore requires not just a strong optimizer, but also robust data versioning, uncertainty-aware stopping rules, and representations that encode synthesis and characterization context along with atomistic structure [33,74,107,194-196].

Reaction-specific application domains

Although methods often transfer across reactions, the target definitions differ enough that reaction-specific patterns have emerged.

For **HER**, machine learning has been most successful in accelerating searches over low-dimensional materials, single-atom motifs, alloy nanoclusters, defect-engineered carbons, and MXene-like systems [85-102,146,149-150,186-192]. In this area, the target is often close to classical descriptor logic: hydrogen adsorption free energy remains dominant, and therefore shallow models with physics-guided descriptors can be highly effective. Many HER studies show that machine learning is most useful not when it replaces physical understanding, but when it systematizes physically meaningful screening over huge composition spaces [2,78-81,85-102].

For **OER**, the discovery problem is harder because stability and reconstruction matter strongly, especially in acidic environments and for transition-metal oxides or oxyhydroxides. As a result, successful OER studies often combine activity prediction with explicit or implicit treatment of composition complexity, site-specific stability, or active-phase evolution [5,16,103-111,177,181,185]. Recent work on NiFe oxyhydroxides, perovskites, and single-atom OER catalysts illustrates that model accuracy alone is insufficient; one also needs chemistry-aware objectives that account for surface phase behavior under bias [103-111,185]. Machine-learning-guided OER design has moved from proof-of-concept prediction to experimentally validated lead identification, including recent noble-metal-free designs found with pretrained geometric models. ⁵

For **ORR**, machine learning has focused both on four-electron fuel-cell catalysis and on two-electron pathways for H₂O₂ generation. Double-atom and cluster electrocatalysts, atomically precise metal-nitrogen-carbon systems, and high-entropy alloys are recurring target classes [112-116,178-180]. Compared

with HER and OER, ORR studies often emphasize selectivity between competing pathways and the challenge of transferring from atomistically controlled representations to realistic catalyst architectures.

For **CO₂RR**, the literature is especially rich because the design space is large and the selectivity landscape is complicated. Early work focused on adsorption-energy screening and surrogate prediction on metals and alloys [6,25-27,65-67,117-130]. More recent work targets active motifs, Cu-based restructuring, additive-guided deposition, dual-atom motifs, and multicarbon selectivity on dynamically evolving surfaces [82-84,117-131]. CO₂RR has also been a proving ground for closed-loop and knowledge-guided methods, including active learning, knowledge graphs, literature mining, and inverse design [82-84,120-131]. Representative 2023-2026 studies include active-motif machine learning for catalyst selection, multi-view active-site learning, and active learning for acetate-selective Cu systems. ⁶

For **NRR**, the strongest machine-learning studies tend to involve single-atom or dual-atom catalysts, where discovery is again constrained by sparse data and uncertainty about operando mechanisms [132-136]. Here, interpretable models are particularly valuable because false positives are common when the true catalytic state or contamination pathway is not fully captured experimentally. NRR therefore illustrates a broader lesson for the field: the more contentious the reaction mechanism, the less useful purely predictive accuracy becomes without mechanistic consistency.

Across reactions, several catalyst families recur. **Single-atom catalysts** offer local-environment regularity that is amenable to descriptor learning and interpretable graph models [79-81,92-93,99-102,110-116,132-150]. **Dual-atom catalysts** enlarge design space while preserving some motif-centric structure, making them natural targets for interpretable ML [70,118,126,133,148,185]. **High-entropy alloys** dramatically increase compositional complexity and therefore benefit from local-cluster descriptors, active learning, and composition-aware representations [35-36,77,138-145,178-180]. **Low-dimensional systems** such as MXenes, MBenes, dichalcogenides, and defect-rich 2D materials remain favored because they combine rich composition space with tractable atomistic representations [78,88-91,146,186-192].

Challenges and Prospects

The central challenge in this field is not simply data scarcity, but **data mismatch**. Most machine-learning models are trained on DFT data from idealized surfaces, yet the catalysts of practical interest operate under bias, in electrolyte, with dynamic adsorbate coverages, variable pH, and often substantial reconstruction. The gap between these regimes is now widely recognized. OCx24 explicitly frames itself as an attempt to bridge computational predictions and experimental outcomes, while OC25 pushes benchmarking toward electrified interfaces. ⁷ Even so, current benchmark progress should not be mistaken for convergence on the full electrochemical problem.

A related challenge is that the field still overuses **proxy labels**. Adsorption energies, limiting potentials, and simple stability descriptors are extremely useful, but they compress away kinetics, transport, poisoning, local environment fluctuations, and degradation. This compression is scientifically acceptable when the proxy is known to be informative, but it becomes hazardous when a model is optimized primarily to fit the proxy rather than the experimental objective. Future work should therefore distinguish more clearly between proxy-prediction success and discovery success.

Uncertainty quantification remains another bottleneck. In materials discovery, a model that is slightly less accurate but better calibrated can be far more useful in practice than a highly accurate but overconfident model. Electrocatalysis makes this especially acute because candidate evaluation is expensive and because training data often undersample the most promising chemical regions. Benchmarking should therefore include calibration error, top-k precision, discovery yield under budget, and out-of-distribution transfer, not just mean absolute error [28,31-34,75-76,159-160]. The field already has the components for this transition, but not yet a universal reporting culture.

There is also a persistent tension between **interpretability and scale**. Descriptor models remain scientifically useful because they reveal trends, inspire mechanistic hypotheses, and often generalize better in small-data regimes. But large geometric models dominate at benchmark scale. The most promising direction is not to choose one or the other. It is to combine them: use equivariant or multimodal models for expressive representation, and then distill or constrain those models with physically meaningful descriptors, sparse rules, or mechanistic structure [68-72,183-185]. The best future systems will likely look less like black-box regressors and more like layered scientific assistants: geometric encoders, uncertainty heads, symbolic or sparse explanatory modules, and retrieval over literature or prior experiments.

A further priority is **multimodal experimental grounding**. Electrocatalyst behavior depends on what the catalyst becomes under operation, not only on what it was before reaction. That means operando spectroscopy, microscopy, diffraction, electrochemical diagnostics, and even synthesis metadata should be treated as first-class learning inputs rather than as post hoc explanatory evidence [33,74,194-195]. The recent move toward multimodal synchrotron-guided discovery is important because it begins to encode real catalyst state transformations rather than static nominal formulas. 8

Finally, the field is entering a period where **foundation models and autonomous systems** may become practically consequential. The most credible path is not unrestricted generation of catalysts from language alone, but tightly constrained systems that combine pretrained atomistic backbones, calibrated active learning, retrieval over literature and datasets, multimodal experimental evidence, and human oversight [151-162]. In other words, the next frontier is not just bigger models. It is well-designed scientific workflows that make every expensive calculation and every experiment more informative.

Conclusion

Machine learning has already changed electrocatalyst discovery, but the nature of that change is more subtle than simple acceleration. The most important contribution of machine learning has been to reframe discovery as an information-allocation problem. Instead of uniformly sampling chemical space, researchers now learn where to compute, where to measure, which active-site motifs to prioritize, which compositions to synthesize, and which descriptor families or representations are worth carrying forward. In this sense, the field has progressed from data fitting to scientific decision support.

Methodologically, several conclusions stand out. First, descriptor-based models remain highly valuable in low-data and interpretability-centered settings, particularly for HER, OER, and motif-centric single-atom or dual-atom catalyst studies. Second, geometric deep learning and machine-learned interatomic potentials have become indispensable for adsorption- and force-related tasks at benchmark scale. Third, active learning and Bayesian optimization are not optional extras but core technologies for managing expensive evaluation budgets. Fourth, multimodal learning and experiment-computation integration are likely to determine whether machine learning can move from benchmark success to real electrochemical impact.

Fifth, inverse design is becoming technically plausible, but it will only be scientifically meaningful when paired with electrochemical realism and explicit uncertainty control.

The most promising research agenda is therefore integrative. Future electrocatalyst discovery systems should combine physically grounded representations, uncertainty-aware learning, benchmarked geometric models, operando-aware experimental data, and closed-loop optimization. If that integration succeeds, machine learning will not merely screen electrocatalysts faster. It will help define a new epistemology of catalyst discovery in which models, simulations, and experiments are co-designed around the objective of discovering robust, selective, scalable materials for sustainable energy conversion.

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